

SMART PARKING ANALYSIS

SQL + Power BI Industry Project

Parking Operations • Revenue • Customer Behavior • Occupancy • Demand

PROJECT	Smart Parking Analysis
INDUSTRY	Smart Parking / Transportation
DATABASE	MySQL
ETL	Power Query
BI PLATFORM	Microsoft Power BI
ANALYTICS	SQL + DAX
DATASET	10,000 parking transaction records
DASHBOARDS	5 interactive pages
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This report documents the actual Power BI dashboard developed for the Smart Parking Analysis project.

01 Project Overview

Smart Parking Analysis is an end-to-end business intelligence project that converts parking transaction data into actionable operational and financial insights. The solution combines MySQL for structured storage, SQL for validation and analysis, Power Query for ETL, Power BI for data modeling and visualization, and DAX for analytical measures.

The completed Power BI report contains five interactive pages covering executive performance, parking operations, revenue, customer and vehicle behavior, and capacity and demand.

02 Industry & Business Problem

Industry: Smart Parking / Transportation Analytics

Parking operators need to understand how facilities are being used, when demand peaks, which facilities generate revenue, how customers and vehicles behave, and whether payment performance creates revenue risk. The dashboard provides a centralized view to support operational planning and data-driven decisions.

Key business challenges

- Uneven demand and occupancy across parking facilities.
- Limited visibility into peak parking hours and time-of-day patterns.
- Need to compare revenue with facility utilization.
- Need to understand customer and vehicle mix.
- Need to monitor payment success, pending and failed transactions.
- Need to identify high-occupancy and underutilized periods.

03 Project Objectives

- Create a structured MySQL database for parking transactions.
- Validate the dataset and perform SQL-based quality checks.
- Clean and transform the data using Power Query.
- Create a Date dimension and one-to-many analytical relationship.
- Build DAX measures for operational, revenue and occupancy KPIs.
- Develop five interactive Power BI dashboard pages.
- Convert dashboard findings into business insights and recommendations.

04 Dataset Description

The project uses a 10,000-row simulated Smart Parking transaction dataset with 14 columns. The data covers transaction identity, parking facilities, vehicle and customer categories, entry timing, duration, payment, pricing, revenue and occupancy.

FIELD	TYPE	DESCRIPTION
Transaction_ID	VARCHAR	Unique transaction identifier
Facility_ID	VARCHAR	Parking facility identifier
Vehicle_Type	VARCHAR	Bike, Car, EV Car or SUV

FIELD	TYPE	DESCRIPTION
Customer_Type	VARCHAR	Member, Regular, Visitor or unknown
Entry_Date	DATE	Date of parking entry
Entry_Hour	INT	Hour of entry, 0–23
Parking_Duration_Min	INT	Parking duration in minutes
Payment_Method	VARCHAR	Payment method
Payment_Status	VARCHAR	Paid, Pending or Failed
Parking_Type	VARCHAR	Parking category
Hourly_Rate	DECIMAL	Hourly parking rate
Discount_Amount	DECIMAL	Applied discount
Net_Revenue	DECIMAL	Final transaction revenue
Occupancy_Rate	DECIMAL	Occupancy rate

05 End-to-End Workflow

01	Data Collection	10,000-row CSV dataset
02	MySQL	Database + parking_transactions table
03	SQL	Validation + business analysis
04	Power BI	MySQL connection
05	Power Query	Cleaning + transformation
06	Data Model	Date dimension + relationship
07	DAX	Measures + KPIs
08	Dashboard	Five Power BI pages
09	Insights	Recommendations + decisions

06 MySQL & SQL Analysis

The CSV dataset was imported into MySQL and stored in the parking_transactions table. SQL scripts support database creation, table creation, data validation and business analysis.

SQL techniques demonstrated

- CREATE DATABASE and CREATE TABLE.
- SELECT, WHERE, GROUP BY, HAVING and ORDER BY.
- COUNT, SUM, AVG and other aggregate functions.
- Subqueries and Common Table Expressions (CTEs).
- Window functions and ranking.
- Date-based monthly analysis.
- Duplicate, NULL and invalid-value validation.

Data architecture: MySQL acts as the source database. Power BI connects to the SQL database rather than importing the original CSV directly.

07 Power Query ETL & Data Cleaning

- Changed columns to correct data types.
- Removed duplicate transaction records.
- Handled missing categorical values.
- Trimmed and cleaned text fields.
- Validated revenue, duration, hourly rate and occupancy.
- Created Entry_Hour_Category: Morning, Afternoon, Evening and Night.
- Created Parking_Duration_Category for duration analysis.
- Created Occupancy_Category for utilization analysis.

Validation checks: duplicate IDs, NULL values, invalid duration, invalid rate, negative revenue, invalid discount, invalid occupancy, invalid entry hour and future dates.

08 Power BI Data Model

A dedicated Date dimension was created in DAX and connected to the parking transaction table using Entry_Date.

Relationship: Dim_Date[Date] → Smartparking analysis[Entry_Date] | Cardinality: 1:* | Cross-filter: Single

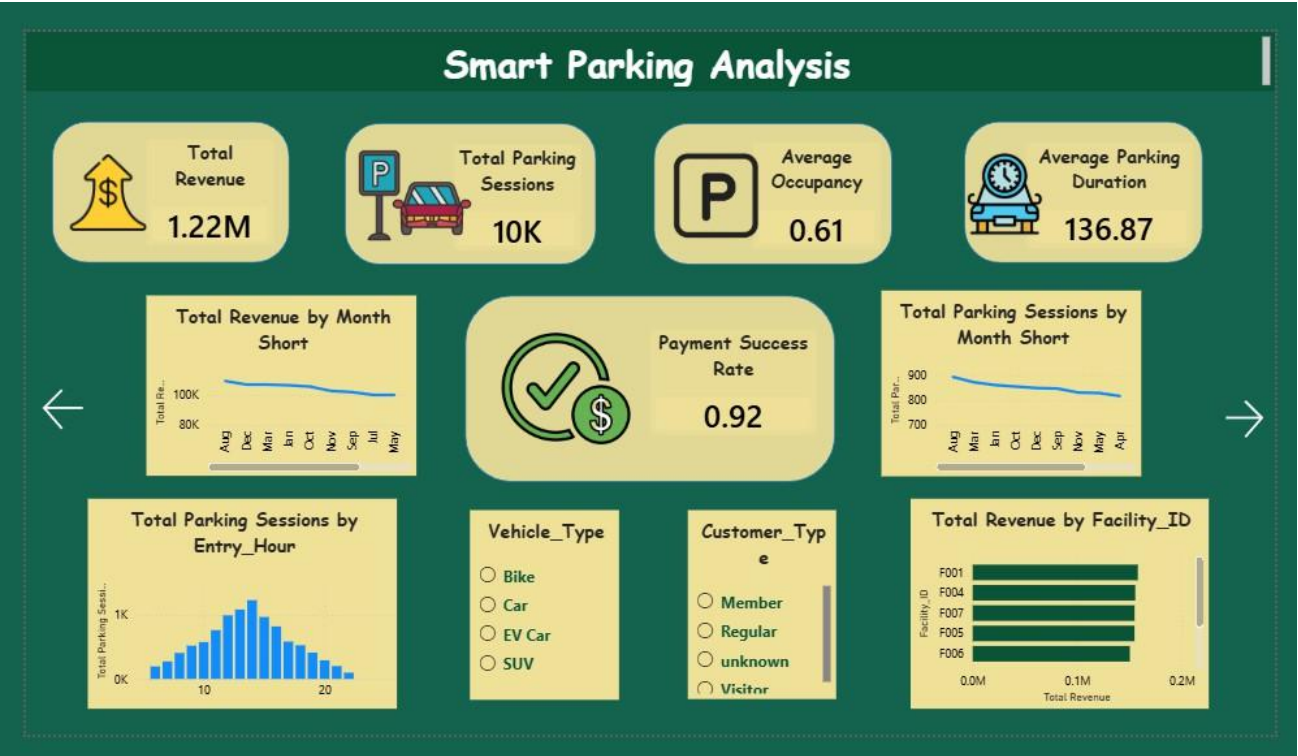
09 DAX Measures & KPI Layer

MEASURE	PURPOSE
Total Revenue	Total Net_Revenue
Total Parking Sessions	Total transaction count
Average Parking Duration	Average stay duration
Average Occupancy	Average occupancy rate
Average Revenue per Session	Revenue efficiency
Paid Sessions	Paid transaction count
Payment Success Rate	Paid sessions / total sessions
Total Discount	Total discount value
Paid Revenue	Revenue from paid transactions
Revenue per Parking Minute	Revenue efficiency by duration
Peak Parking Hour	Highest-demand entry hour
Peak Occupancy	Maximum observed occupancy
High / Low Occupancy	Occupancy risk segmentation

10 Actual Power BI Dashboard Screenshots

These are the actual dashboard screenshots supplied for the project – not generated mockups.

01 Executive Overview



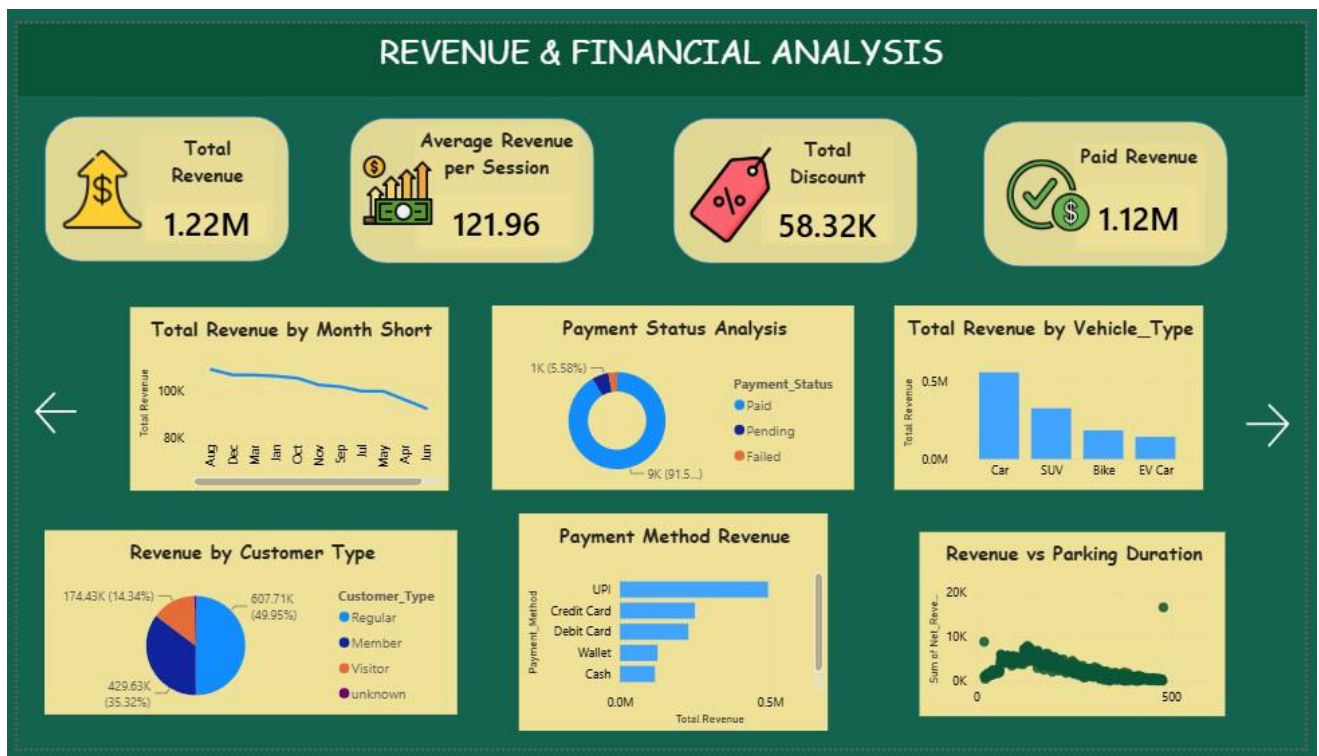
Actual Power BI dashboard screenshot.

02 Parking Operations



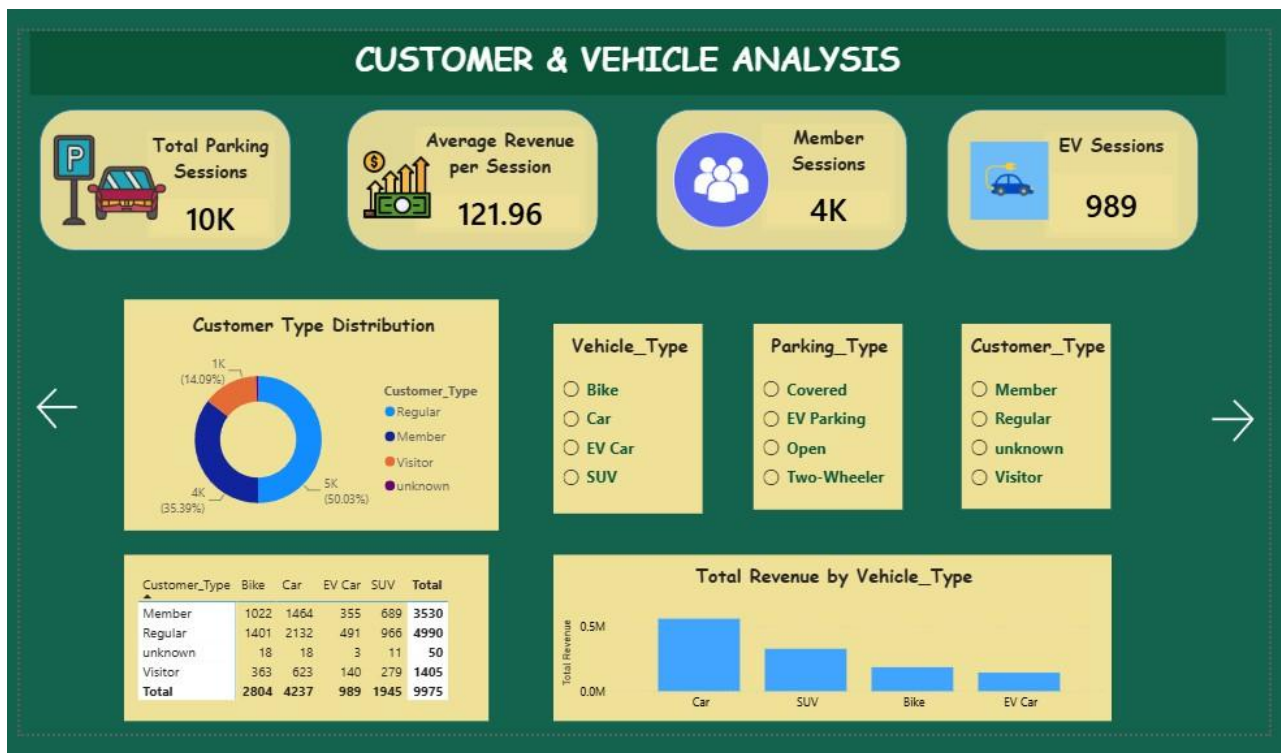
Actual Power BI dashboard screenshot.

03 Revenue & Financial Analysis



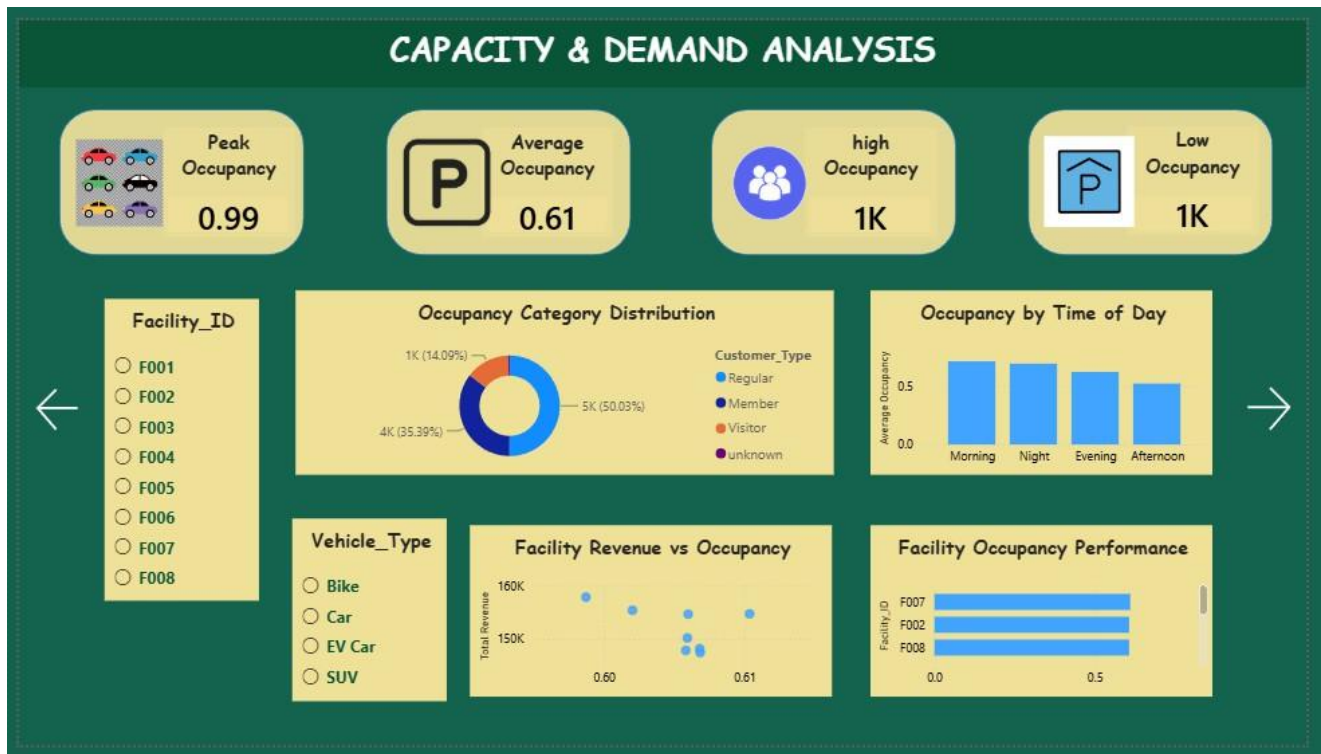
Actual Power BI dashboard screenshot.

04 Customer & Vehicle Analysis



Actual Power BI dashboard screenshot.

05 Capacity & Demand Analysis



Actual Power BI dashboard screenshot.

11 Dashboard KPI Snapshot

KPI	VALUE SHOWN
Total Revenue	1.22M
Total Parking Sessions	10K
Average Occupancy	0.61
Average Parking Duration	136.87 minutes
Payment Success Rate	0.92
Average Revenue per Session	121.96
Paid Revenue	1.12M
Total Discount	58.32K
Peak Occupancy	0.99
EV Sessions	989

Note: Power BI displays some values using compact formats such as 10K and 1.22M. The underlying dataset contains 10,000 transaction records.

12 Business Questions Answered

- Which facilities generate the highest revenue?
- Which facilities have the highest occupancy?
- What are the peak parking hours?
- How does demand vary by time of day?
- Which vehicle types contribute the most sessions and revenue?
- Which customer segment contributes the most revenue?
- What is the average parking duration?
- Which payment methods generate the most revenue?
- Where are high-occupancy and low-occupancy periods?
- How does facility revenue compare with occupancy?

13 Business Insights

- The Executive Overview reports total revenue of 1.22M and approximately 10K parking sessions.
- Average occupancy is 0.61, while the Capacity & Demand page shows a peak occupancy of 0.99.
- Average parking duration is 136.87 minutes, making duration an important operational factor.
- Payment Success Rate is 0.92 and Paid Revenue is 1.12M.
- Car is the largest vehicle category by sessions and revenue in the supplied dashboard.
- Member sessions represent a major customer segment, while Regular customers also contribute strongly to revenue.

- The Operations page identifies 14 as the peak parking hour in the supplied dashboard.
- The Capacity & Demand page highlights high-occupancy and low-occupancy segments for capacity planning.

14 Recommendations

- Optimize space allocation at consistently high-occupancy facilities.
- Investigate low-occupancy facilities using targeted promotions and pricing.
- Align staffing and entry/exit resources with peak parking hours.
- Use vehicle and customer mix to improve parking-space planning.
- Monitor pending and failed payments to reduce potential revenue leakage.
- Use occupancy and demand patterns as inputs for future dynamic pricing.
- Continue monitoring EV demand to support future EV parking and charging infrastructure.

15 GitHub Deliverables

ITEM	PURPOSE
README.md	Project overview, workflow and dashboard preview
Dataset/	10,000-row Smart Parking dataset
SQL/	Database, table, validation and analysis scripts
PowerBI/	Smart_Parking_Analysis.pbix
Screenshots/	Actual five dashboard screenshots
Documentation/Project_Report.pdf	Final project report
Documentation/Data_Dictionary.xlsx	Dataset documentation

16 Future Enhancements

- Integrate real-time parking sensor / IoT data.
- Add parking slot-level availability.
- Add Customer_ID for repeat-customer analysis.
- Add location/GPS analytics.
- Develop dynamic pricing recommendations.
- Analyze EV charging utilization.
- Add predictive parking-demand modeling.
- Publish the report through Power BI Service.

17 Conclusion

Smart Parking Analysis demonstrates a complete analytics lifecycle from raw parking data to business intelligence. The project combines MySQL, SQL, Power Query, Power BI data modeling, DAX and interactive visualization to support parking operations, revenue analysis, customer understanding, occupancy monitoring and demand planning.